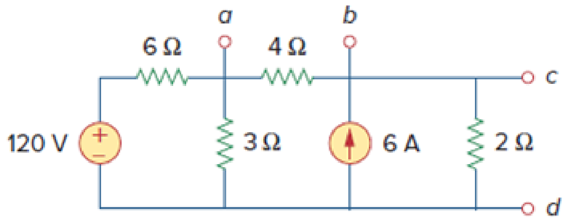
Problem

Given the circuit in Fig. 4.117, obtain the Norton equivalent as viewed from terminals:

(a) *a-b*

(b) *c-d*

Figure. 4.117



Step-by-step solution

Step 1 of 6

(a)

Find Norton equivalent circuit viewed from a and b terminals.

Find Norton resistance by short circuit the voltage source and open circuit the current source.

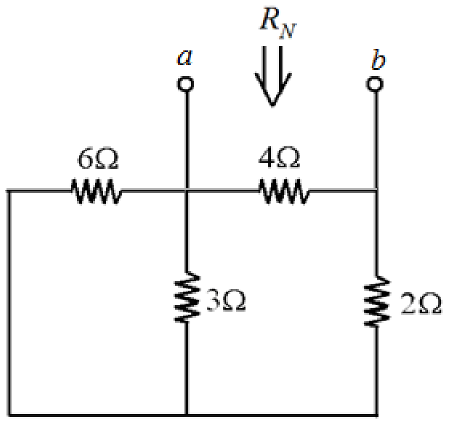


Figure 1

Norton resistance is,

$$R_N = 4\Omega \,||\, \left(2\Omega + \left(6\Omega \,||\, 3\Omega\right)\right)$$

$$R_N = 4 \,||\, \left(2 + \left(\frac{6\times 3}{6+3}\right)\right)$$

$$R_N = 4 \,||\, (2+2)$$

$$R_N = \frac{4\times 4}{4+4}$$

Norton resistance viewed form terminals a and b is, $\boxed{R_N = 2\Omega}$.

Step 2 of 6

Find Norton current.

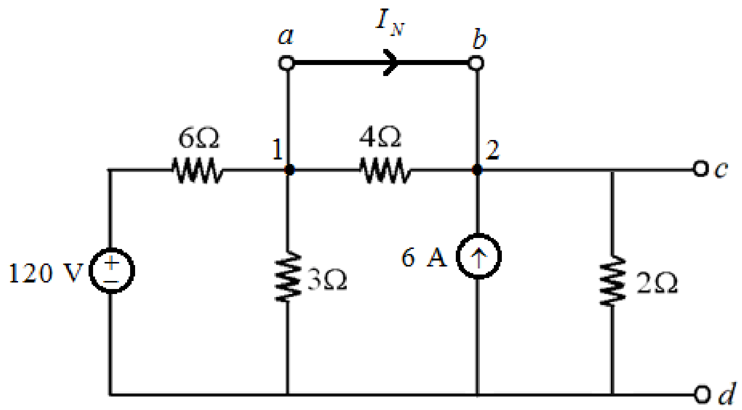


Figure 2

Observe that both

Apply KCL at node 1.

$$\frac{v_1}{3}+\frac{v_1-120}{6}+I_N=0 \quad \ldots\ldots (1)$$

Observe that nodes 1 and 2 are short circuited. Thus,

$$v_1=v_2$$

Apply KCL at node 2.

$$I_N+6=\frac{v_2}{2}$$

Substitute v_1 for v_2 in the equation.

$$I_N=\frac{v_1}{2}-6$$

Substitute $\frac{v_1}{2}-6$ for I_N in equation (1).

$$\frac{v_1}{3}+\frac{v_1-120}{6}+\frac{v_1}{2}-6=0$$

$$v_1\left(\frac{1}{3}+\frac{1}{6}+\frac{1}{2}\right)=6+\frac{120}{6}$$

$$v_1\left(\frac{2+1+3}{6}\right)=6+20$$

$$v_1=26\text{ V}$$

Step 3 of 6

Substitute 26 V for v_1 in equation (1).

$$\frac{v_1}{3}+\frac{v_1-120}{6}+I_N=0$$

Substitute 26 for v_1 in the equation.

$$\frac{26}{3}+\frac{26-120}{6}+I_N=0$$

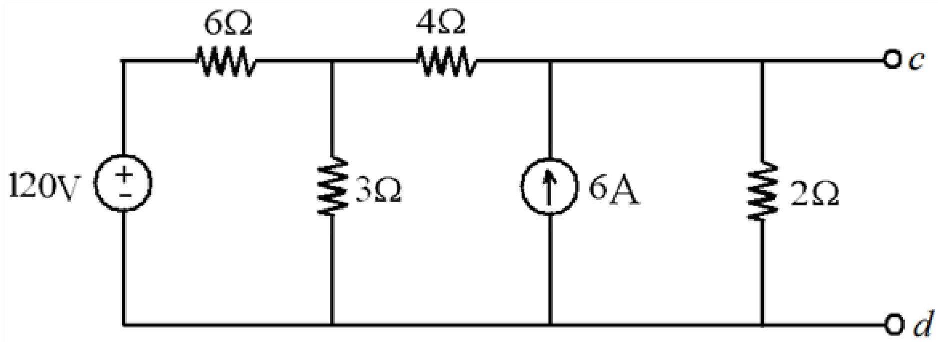
$$I_N=7\text{ A}$$

Thus, Norton current viewed from terminals a and b is, $\boxed{I_N=7\text{ A}}$.

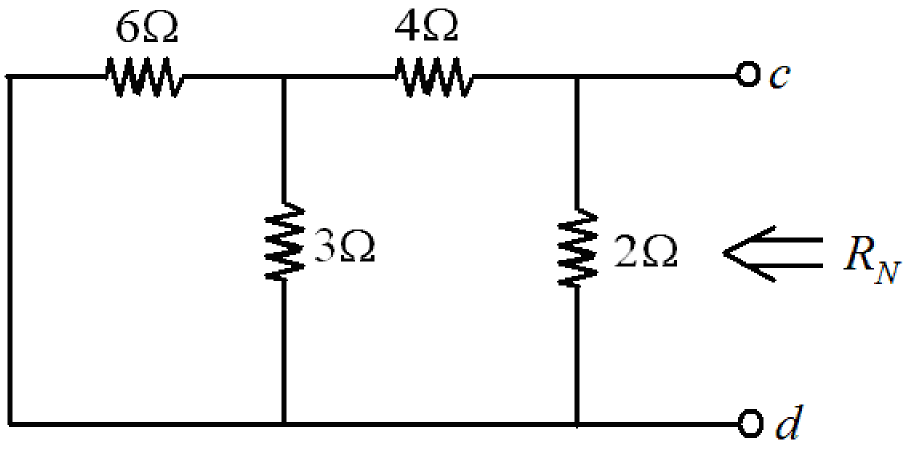
Step 4 of 6

(b)

Find the Norton equivalent viewed from terminals c and d.



Find the Norton resistance viewed from terminals c and d.



Norton resistance is,

$$R_N = \left[(6 \| 3) + 4\right] \| 2$$

$$R_N = \left[2 + 4\right] \| 2$$

$$R_N = 6 \| 2$$

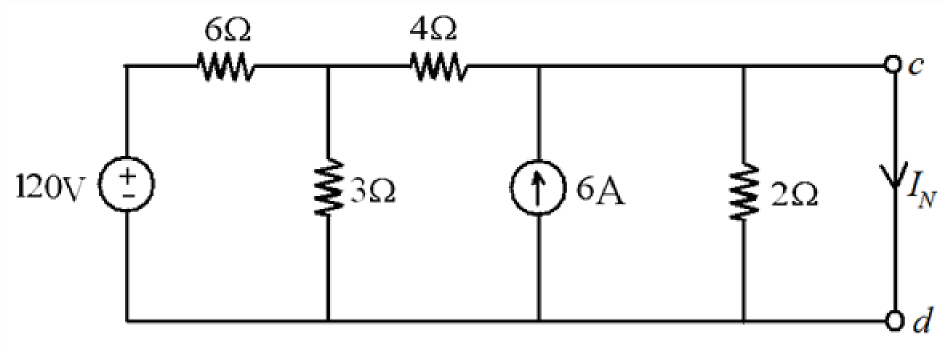
$$R_N = \frac{6 \times 2}{6 + 2}$$

$$\boxed{R_N = 1.5\Omega}$$

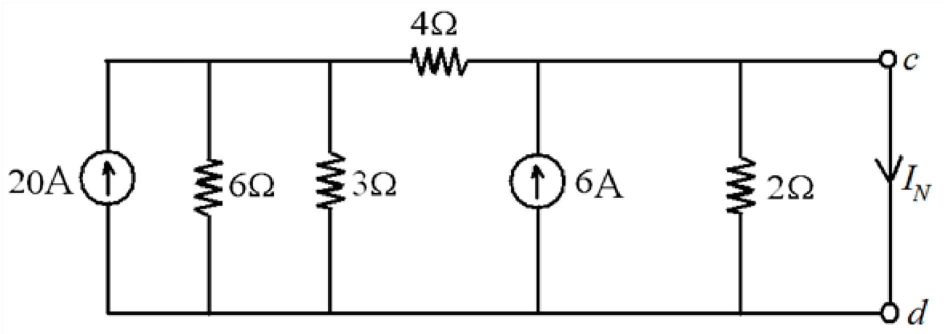
Norton resistance viewed form terminals c and d is, $\boxed{R_N = 1.5\ \Omega}$.

Step 5 of 6

Find Norton's current.



Apply source transformation to convert the 120 V and 6Ω resistor into current source in parallel with the resistor.

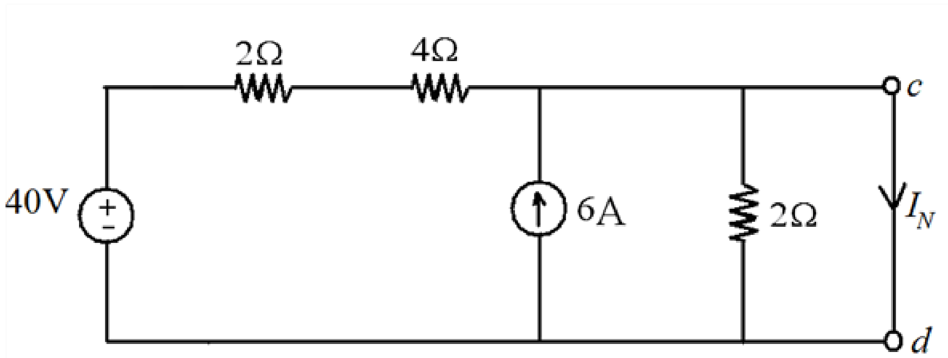


In this $6\ \Omega$ and $3\ \Omega$ are in parallel

$$6\Omega \| 3\Omega = \frac{6\times 3}{6+3}$$
$$6\Omega \| 3\Omega = 2\Omega$$

Replace the $6\ \Omega$ and $3\ \Omega$ with the equivalent 2Ω resistor.

Now, apply source transformation to convert the 20 A and the $2\ \Omega$ resistor into voltage source in series with the resistor.



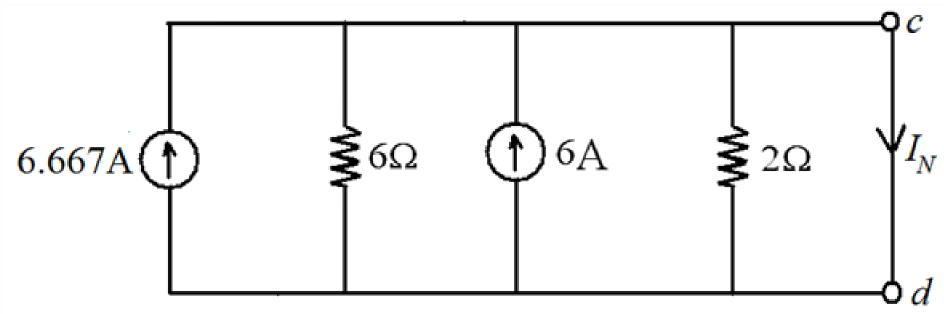
Resistors $2\ \Omega$ and $4\ \Omega$ are in series.

$$2\ \Omega + 4\ \Omega = 6\ \Omega$$

Step 6 of 6

Replace the $2\,\Omega$ and $4\,\Omega$ with the equivalent $6\,\Omega$ resistor.

Now, apply source transformation to convert the voltage source 40 V and resistor $6\,\Omega$ into current source in parallel with the resistor.



From the circuit, due to short-circuit terminals across c and d, the total current generated from the sources flow through the short circuit terminals c and d. No current flow through $6\,\Omega$ and $2\,\Omega$ resistors.

$I_N = 6 + 6.667$

$\boxed{I_N = 12.667\text{A}}$.